PROBLEM 03

Kinematics, Astroid

Original statement · Problems.pdf, p. 6

Intrinsic Kinematics

A point particle describes the astroid:

$$x = a\cos^3\phi, \qquad y = a\sin^3\phi$$

with a motion whose tangential acceleration is constant, k. If the point is initially at rest at $\phi = 0$, compute the evolution of the position as a function of time while the particle is in one of the smooth segments of the astroid. *(Hint:* $ds = \sqrt{dx^2 + dy^2}$*).*

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** a particle P moves on the astroid $x = a\cos^3\phi, y = a\sin^3\phi$ with constant tangential acceleration $a_t = k$ (the statement calls it k) and starts from rest at $\phi = 0$.
- **Hypotheses:** $a > 0$, $k > 0$. The motion stays within one smooth segment of the curve, here the first-quadrant arc $0 \le \phi \le \pi/2$ (the astroid has cusps at $\phi = 0, \pi/2, \pi, 3\pi/2$). Starting at the cusp $\phi = 0$ the particle could enter either neighbouring arc; we choose the one with ϕ increasing, so that s increases with ϕ and $d\phi \ge 0$.
- **Constraints and degrees of freedom:** the curve is a holonomic, scleronomic constraint in the plane, so $\mathrm{CDOF} = 2 - 1 = 1$. The coordinate is ϕ, or the arc length s measured from $\phi = 0$.
- **Parameters and units:** scale a [m], tangential acceleration k [m/s^2], time t [s].

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ with the curve in the Oxy plane.
- **Parametric curve:** $x(\phi) = a\cos^3\phi$, $y(\phi) = a\sin^3\phi$. Differentiating with respect to the parameter (chain rule):

$$\frac{dx}{d\phi} = 3a\cos^2\phi\,(-\sin\phi) = -3a\cos^2\phi\sin\phi, \qquad \frac{dy}{d\phi} = 3a\sin^2\phi\cos\phi$$

PHASE 3

Derivation

3.1 Kinematics

Pedagogical Justification: we use the arc length as the coordinate along the path. The differential arc length follows from the hint, $ds = \sqrt{dx^2 + dy^2}$:

$$ds^2 = \left(-3a\cos^2\phi\sin\phi\right)^2 d\phi^2 + \left(3a\sin^2\phi\cos\phi\right)^2 d\phi^2 = 9a^2\cos^4\phi\sin^2\phi\,d\phi^2 + 9a^2\sin^4\phi\cos^2\phi\,d\phi^2$$

$$ds^2 = 9a^2\cos^2\phi\sin^2\phi\left(\cos^2\phi + \sin^2\phi\right)d\phi^2 = 9a^2\cos^2\phi\sin^2\phi\,d\phi^2$$

For $0 \le \phi \le \pi/2$ both $\sin\phi$ and $\cos\phi$ are non-negative (and $d\phi \ge 0$), so the positive root is

$$ds = 3a\sin\phi\cos\phi\,d\phi$$

- **Tangential acceleration.** The tangential component of the acceleration is the second derivative of the arc length: $a_t = \ddot{s} = k$.

3.2 Forces

Not applicable: kinematics-only problem. The tangential acceleration is given.

3.3 Equations of Motion

Not applicable: kinematics-only problem. The law $\ddot{s} = k$ is the statement itself.

3.4 Integration and Answers

Step 1: arc length as a function of time

> **Pedagogical Justification:** integrate $\ddot{s} = k$ twice, with the initial conditions $\dot{s}(0) = 0$ (at rest) and $s(0) = 0$ (at $\phi = 0$):

$$\dot{s}(t) = \int_0^t k\,dt' = [kt']_0^t = kt - 0 = kt, \qquad s(t) = \int_0^t kt'\,dt' = \left[\frac{1}{2}kt'^2\right]_0^t = \frac{1}{2}kt^2 - 0 = \frac{1}{2}kt^2$$

Step 2: arc length as a function of ϕ

> **Pedagogical Justification:** integrate ds from $\phi = 0$. With the substitution $w = \sin u$, $dw = \cos u\,du$, the limits become $w = 0$ at $u = 0$ and $w = \sin\phi$ at $u = \phi$:

$$s(\phi) = \int_0^{\phi} 3a \sin u \cos u\,du = 3a \int_0^{\sin\phi} w\,dw = 3a\left[\frac{w^2}{2}\right]_0^{\sin\phi} = 3a\left(\frac{\sin^2\phi}{2} - 0\right) = \frac{3a}{2}\sin^2\phi$$

Step 3: equating both expressions

> **Pedagogical Justification:** the distance travelled along the curve is the same quantity in both descriptions, $s(t) = s(\phi)$:

$$\frac{3a}{2}\sin^2\phi = \frac{1}{2}kt^2 \implies \sin^2\phi = \frac{kt^2}{3a} \implies \sin\phi = \sqrt{\frac{k}{3a}}\,t$$

where the positive root is taken because $0 \le \phi \le \pi/2$. Therefore

$$\boxed{\phi(t) = \arcsin\left(\sqrt{\frac{k}{3a}}\, t\right)}$$

Step 4: position as a function of time

Pedagogical Justification: with $\sin\phi = \sqrt{k/(3a)}\, t$ and $\cos\phi = \sqrt{1-\sin^2\phi} = \sqrt{1 - kt^2/(3a)}$ (non-negative in the first quadrant):

$$\boxed{x(t) = a\cos^3\phi = a\left(1 - \frac{kt^2}{3a}\right)^{3/2}}, \qquad \boxed{y(t) = a\sin^3\phi = a\left(\frac{k}{3a}\right)^{3/2} t^3}$$

These hold for $0 \le t \le t_{\text{cusp}}$, where t_{cusp} is defined in Phase 4.

PHASE 4

Interpretation & Checks

- **Dimensional check:** $\left[\sqrt{\frac{k}{3a}}\, t\right] = \sqrt{\frac{\mathrm{m\,s^{-2}}}{\mathrm{m}}}\,\mathrm{s} = \mathrm{s^{-1}}\cdot\mathrm{s} = 1$, so the argument of arcsin is dimensionless, as it must be.
- **Physical limits:** at $t = 0$, $\phi = 0$ and the particle is at the cusp $(a, 0)$. It reaches the next cusp, $\phi = \pi/2$ at $(0, a)$, when $\sin\phi = 1$:

$$\sqrt{\frac{k}{3a}}\, t_{\text{cusp}} = 1 \implies t_{\text{cusp}} = \sqrt{\frac{3a}{k}}$$

The total length of the arc is $s(\pi/2) = 3a/2$, consistent with $\frac{1}{2}kt_{\text{cusp}}^2 = \frac{1}{2}k\cdot\frac{3a}{k} = \frac{3a}{2}$. The solution stops at the cusp for a physical reason. The tangent is $\mathbf{e}_t = (-\cos\phi, \sin\phi)$ and $d\mathbf{e}_t/d\phi = (\sin\phi, \cos\phi)$ has unit norm, so $\rho = \dfrac{ds/d\phi}{\|d\mathbf{e}_t/d\phi\|} = 3a\sin\phi\cos\phi$, which tends to 0 at both cusps, with the centre of curvature on the normal $\mathbf{e}_n = (\sin\phi, \cos\phi)$ (the arc is convex toward the origin). At arrival the speed is $\dot{s}(t_{\text{cusp}}) = kt_{\text{cusp}} = \sqrt{3ak} \neq 0$, so $a_n =$

$\dot{s}^2/\rho \to \infty$: the motion cannot continue with constant a_t through the cusp. At the start, $a_n \approx \dfrac{(kt)^2}{3a\phi} \to 0$ because $\phi \approx \sqrt{k/(3a)}\,t$, so the departure cusp is regular. Explicitly, $a_n \approx \dfrac{k^2t^2}{3a\sqrt{k/(3a)}\,t} = k\sqrt{\dfrac{k}{3a}}\,t \to 0$.

- **Check against the official key:** the key of `Problems.pdf` is $\phi = \arcsin\left(\sqrt{k/(3a)}\,t\right)$ and $s = kt^2/2$ (read on the rendered page), which equals the result above. Sympy also confirms that $s(\phi(t)) - kt^2/2 = 0$ identically.

Source: Problems.pdf